



TADS: a novel dataset for road traffic accident detection from a surveillance perspective

Yachuang Chai¹ · Jianwu Fang² · Haoquan Liang³ · Wushouer Silamu^{1,4}

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Abstract

With the continuous development of socio-economics, the rapid increase in the use of road vehicles has led to increasingly severe issues regarding traffic accidents. Timely and accurate detection of road traffic accidents is crucial for mitigating casualties and alleviating traffic congestion. Consequently, road traffic accident detection has become a focal point of research recently. With the assistance of advanced technologies such as deep learning, researchers have designed more accurate and effective methods for detecting road traffic accidents. However, deep learning models are often constrained by the scale and distribution of their training datasets. Presently, datasets specifically tailored for road traffic accident detection suffer from limitations in scale and diversity. Furthermore, influenced by the recent surge in research on intelligent driver assistance systems, datasets from the surveillance perspective (the third-person viewpoint) are fewer than those from the driver's perspective (the first-person viewpoint). Considering these shortcomings, this paper proposes a new dataset, Traffic Accident Detection from the Perspective of Surveillance (TADS). To the best of our knowledge, we are the first to attempt to detect traffic accident under the surveillance perspective with the aid of eye-gaze data. Leveraging the special data components within this dataset, we design the RF-RG model (input: the RGB and optical flow values of the frames; output: the RGB and gaze values of the predicted frame) for detecting road traffic accidents from a surveillance perspective. Comparative experiments and analyses are conducted with existing major detection methods to validate the efficacy of the proposed dataset and the approach. The TADS dataset has been made available at: <https://github.com/cyc-gh/TADS/>.

Keywords Traffic accident detection · Video anomaly detection · Surveillance video · Deep learning

1 Introduction

With the continuous development of society, the number of road vehicles has increased rapidly, leading to a significant rise in road traffic accidents, which have become a major concern impeding societal progress and jeopardizing the safety of lives and properties. Addressing the frequent occurrence of traffic accidents is crucial for the timely response to such incidents, involving actions like promptly blocking relevant roads, dispatching medical and firefighting rescue personnel, and analyzing the causes of accidents. These measures are essential for the timely rescue of accident victims and the management of the scene to minimize secondary accidents and collateral damage.

As surveillance cameras proliferate on roads, road monitoring has emerged as a primary means of monitoring traffic accidents. However, the challenge lies in the real-time processing and timely response to the massive amount of surveillance videos generated by these cameras. The automation level of detecting and analyzing traffic accidents in these extensive data remains a subject requiring further attention. Therefore, leveraging intelligent technology to assist in the detection, warning, and analysis of traffic accidents holds practical significance. Accurate and timely detection of traffic accidents on roads is crucial for the prompt rescue of casualties and for alleviating traffic congestion.

To this end, an increasing number of researchers have applied deep learning methods, which have been a popular and effective way in recent years, to accident detection, yielding some notable results. Among these methods, most of them are inspired by video anomaly detection. But it is noteworthy that there exists a close correlation and slight difference between road traffic anomalies and road traffic accidents according to reference [1], in which road traffic anomalies encompass more scenarios and longer time windows compared to road traffic accidents. This paper will not delve into the detailed differences between them. However, for the advancement of research on road traffic accident detection, the objective of this study is to detect time windows of road traffic anomalies that include road traffic accident scenes. Thus, the term "road traffic accident detection" in this paper specifically refers to detecting time windows of road traffic anomalies containing road traffic accident scenes.

Most existing deep learning detection methods heavily rely on training data; a large-scale, high-fidelity dataset is crucial for improving the detection performance of these deep models. However, the current available datasets for road traffic accident detection still suffer from the following shortcomings:

1. Limited variety and small scale of open-source datasets specifically designed for road traffic accident detection under the surveillance perspective. The goal of detecting road traffic accidents is to identify occurrences or conclusions of accidents from video data and mark the time and spatial positions of accidents in the video. Therefore, datasets for road traffic accident detection need to include a sufficient number of traffic accident-relevant video segments. Due to the high hardware requirements and acquisition costs for video data collection, coupled

with the rarity of road traffic accidents in daily video segments, existing datasets for road traffic accident detection are generally small in scale and lack diversity in accident category, which would hinder the performance of the deep models. [1] presents some available datasets for road accident detection and their characteristics, as shown in Table 1.

As can be seen from Table 1, the existing open-source datasets that can be used for road traffic accident detection are not abundant enough compared to other computer vision tasks, and most of them only contain limited video segments or frames.

- Existing open-source datasets for road traffic accident detection mostly focused on the driver's perspective, with fewer datasets catering to the surveillance perspective. In addition, these datasets mainly comprised three viewpoints: Bird's Eye View (BEV), Dashcam, and Surveillance. However, the majority of them are from the dashcam perspective, refer to Table 1. There is a scarcity of surveillance perspective accident datasets captured in real-world scenarios with a considerable scale. For instance, as illustrated in Table 1, among the available open-source datasets for road traffic accident detection, only five are oriented toward the surveillance perspective, with merely three being non-synthetic. Among these three

Table 1 Some available datasets for road accident detection and their characteristics, with the attributes of the number of sequences (Seq. num), annotations, synthetic (S) or real (R), and observation views (dashcam, surveillance, and BEV). Refer to Table 4 in [1]

Datasets	Seq. num	Annotations	S/R	Observation view
A3D [2]	1500	T	R	Dashcam
DVAD [3]	60	T, Text	R	Dashcam
Drive-Anomaly106 [4]	106	T, S	R	Dashcam
RetroTrucks [5]	474	T, S	R	Dashcam
TNAD [6]	106	T, S	R	BEV
ADV [7]	100	T, S	R	Dashcam
TAD-1 [8]	500	T, S	R	Dashcam, Surveillance
TaskFix [9]	1436F ¹	T, Att	R	Surveillance
USDC [10]	122	T, S	R	Dashcam
Traffics [11]	29	T, S	R	Surveillance
DADA-2000 [12]	2000	T, Text, Att	R	Dashcam
DoTA [13]	4,677	T, S	R	Dashcam
TAD-2 [14]	333	T, S	R	Surveillance
DADA-seg [15]	313	T, SEG	R	Dashcam
MP-RAD [16]	2000	T	S	Surveillance
DRAMA [17]	17,785	Risk, S, Text	R	Dashcam
CTAD [18]	1100	T	S	Surveillance

¹1436F: 1436 frames

datasets, namely TaskFix, Traffics, and TAD-2, they contained only 1436 frames, 29 video segments, and 333 video segments, respectively. Consequently, the datasets that could be used to train deep models and captured from the surveillance viewpoint in the real world are not rich enough.

3. The existing annotated information in open-source datasets for road traffic accident detection is relatively limited. Video datasets contain abundant data information, so rich annotation could facilitate the model to learn the patterns in the dataset better. For video datasets, a large number of video frames and corresponding temporal and spatial annotations per frame will greatly benefit the model in learning tasks related to them, but adding annotation to a large-scale video dataset often brings high costs. Therefore, the annotation information of some datasets available is also not abundant.

In light of these shortcomings, this paper collects and proposes a new dataset named TADS (Traffic Accident Detection from the Perspective of Surveillance). This dataset comprises 966 accident video segments, totaling 259,860 frames, with both temporal and spatial annotations provided for all frames. Temporal annotations include the start of anomalies, the start of accidents, and the end of accidents at the frame level. Spatial annotations incorporate eye-tracking data as a means of indicating areas of interest for human vision, which might aid in accident detection and prediction. Additionally, the dataset includes annotations for diverse attributes about accident, such as accident scenes, weather conditions, lighting, and accident categories for each accident segment.

The contributions of this paper are as follows:

1. We have created a new dataset named TADS (Traffic Accident Detection from the Perspective of Surveillance), which features a large-scale, richly annotated collection tailored for addressing the road traffic accident detection problem from a surveillance perspective.
2. Drawing upon existing research on road traffic accident detection, we have constructed a typical model capable of leveraging the special annotation information present in our dataset. We conducted comparative experiments and evaluations of the detection performance of this model with major existing detection methods on the TADS. The comparisons affirm the usability and effectiveness of the data contained within our dataset for accident detection. Due to the preprocessing requirements for object detection in input data, the methods in this paper conducted accident detection offline for the sake of experimental convenience and fair comparison. However, these methods can also be adapted for online detection in practical applications with slight modifications, such as incorporating object detection preprocessing after receiving video frames online.

The remainder of this paper is organized as follows: Section 2 reviews the related datasets and methods for traffic accident detection. Section 3 presents the process of constructing the TADS dataset, including data collection, organization, annotation, and comparison with related datasets. Section 4, based on the special

annotation information in the TADS dataset, presents the construction of novel detection model. Section 5 provides the extensive experiments and analysis. Section 6 concludes the paper and provides insights into future research directions.

2 Related work

2.1 Available dataset for road traffic accident detection

As shown in Table 1, existing datasets can be categorized into three main classes by the viewpoint perspective: Bird's Eye View (BEV), Dashcam, and Surveillance. Most of the video data in these datasets are collected through the in-vehicle camera or online platforms such as YouTube. Below, we will list several representative datasets:

A3D [2]: This dataset comprises 1500 video segments of road anomalies sourced from YouTube, each containing a traffic anomaly event. The videos contain various vehicle types mainly from East Asia, all captured from the driver's perspective. With a total of 128,175 frames, the dataset encompasses 18 kinds of traffic accidents. Furthermore, the dataset was later expanded by the team to create DoTA, comprising 4677 driver's perspective videos totaling 731,932 frames.

TAD [14]: To address issues such as small-scale, single-scenario, low event counts, and limited event types in existing datasets, TAD consists of 333 videos (261 of which contain traffic accidents), captured from a surveillance perspective. The video resolutions range from (862, 530) to (1432, 990), and for frame-level tasks, TAD offers 24,810 annotated frames covering four major accident categories. Notably, TAD's sources are not solely online platforms; most of the data are from a video analysis platform developed by the team.

CADP [19]: This dataset, compiled from platforms like YouTube, comprises 1416 surveillance perspective videos, all featuring traffic accidents. It serves various tasks such as object detection, object tracking, and collision detection. However, only 205 videos in the dataset have annotations related to spatial and temporal dimensions.

DADA-2000 [12]: Focused on the dashcam viewpoint, DADA-2000 is a dataset designed for predicting driver attention in accident scenarios. It is a collaborative effort between Chang'an University and Xi'an Jiaotong University and consists of 2,000 video segments sourced from platforms like Youku and Tencent. In total, the dataset comprises 658,476 frames and categorizes accidents into 54 different types based on the kinds of participants involved.

SUTD-TrafficQA [20]: Combining online and offline data collection, this dataset includes various perspectives such as driver's and surveillance views, totaling 10,080 videos. However, it primarily targets video question-answering tasks in traffic scenarios, containing 62,535 question-answer pairs related to traffic videos. The dataset aims to enhance models' understanding of traffic scenes in a question-answer way.

2.2 Existing road traffic accident detection models

Currently, methods for detecting road traffic accidents in videos can be mainly categorized into two types: supervised learning-based classification methods and unsupervised video anomaly detection methods.

Supervised learning-based classification methods: With the rise of machine learning techniques and artificial neural networks, their performance in simple classification and regression tasks has become increasingly reliable. So some scholars directly consider the traffic accident detection problem as a classification issue, employing simple network structures or models for detection. Nejd et Dogru [21] et al. utilized vehicle position and velocity data, employing artificial neural networks, support vector machines, and random forest methods for accident detection. The results indicated that the random forest method performed better in terms of detection accuracy. Wei Wang [22] conducted experiments on the I-880 dataset in California using Partial Least Squares Regression (PLSR) and Support Vector Machine (SVM) methods, showing that PLSR had stronger accident detection capabilities. Dipti Srinivasan [23] evaluated the adaptability of Multi-Layer Feedforward Neural Network (MLFNN), Basic Probabilistic Neural Network (BPNN), and Constructive Probabilistic Neural Network (CPNN). They found that MLFNN performed best in accident vs. non-accident classification, but CPNN showed superior performance after adaptive modeling in a new environment. Dewant Katare [24] addressed the vehicle collision warning/avoidance task, employing both supervised training methods for sensor data through classification and regression and conducting experimental analysis using the RTMaps embedded framework.

Unsupervised video anomaly detection methods: Unsupervised video anomaly detection involves processing surveillance videos using computer vision techniques to automatically discover segments that deviate from general patterns. Applying this method to traffic surveillance videos can detect accident segments with abnormal characteristics. Due to advantages such as easy acquisition of training data and low annotation costs in an unsupervised manner, this approach has become a current research focus. For instance, Li W [25] experimented with hierarchical mixed dynamic structure models, conditional random fields, and global consistency reasoning on the UCSD pedestrian anomaly detection dataset. Medel J R [26] created a composite model to extend the ConvLSTM network, applying ConvLSTM to abnormal video segment detection using a rule-based evaluation algorithm, validated on some surveillance perspectives datasets, like UCSD [27], Avenue [28], and Subway [29]. Luo W [30] utilized Temporally Coherent Sparse Coding (TSC) for anomaly detection, speeding up the anomaly prediction process using stacked Recurrent Neural Networks (sRNN). Additionally, they constructed the ShanghaiTech Campus dataset with various monitoring angles. Liu W [31], utilizing the spatial-temporal features of consecutive video frames, employed adversarial training to predict future video frames. Frames with a significant deviation from the predicted frames were determined as anomalous, and the effectiveness of this method was evaluated on several commonly used surveillance video datasets. Chong Y S [32] used ConvLSTM to extract spatial-temporal features of videos, to train an autoencoder on normal video segments, and applied it to detect abnormal videos, demonstrating

the advantages of the unsupervised detection method. Moreover, in surveillance perspectives, accident or anomalous areas often occupy a small spatial region in video frames. Therefore, some methods address this detection challenge by first performing objects detection on important objects in video frames before accident or anomaly discrimination. For example, Liu Z [33] initially detected important objects in video frames using popular objects detection methods, reconstructed the optical flow of target objects in normal data part using a memory autoencoder, and used this reconstructed optical flow as a condition to help future frame prediction. This approach, combining objects detection, frame reconstruction, and frame prediction, achieved favorable results in video anomaly detection benchmarks. Objects detection in the preprocessing stage can help the model ignore fixed backgrounds with little relevance in surveillance perspectives, focus on learning important objects with motion changes, and improve training speed and the pertinence of learning. Regarding the advantages of the approaches above, this study followed the detection way in [33], conducting objects detection preprocessing on the dataset.

Due to the insufficiency of datasets for road traffic accident detection under surveillance perspectives, currently, methods for comparison in road traffic accident detection tasks have a broad scope and lack specificity in traffic accidents learning. For example, as road traffic accidents share some similarities with video anomalies to some extent, methods related to video anomaly detection are often used for comparisons. However, abnormal videos are not equivalent to road traffic accidents absolutely. It is crucial to enhance the model's unique understanding of road traffic accidents on a benchmark with a considerable number of road traffic accident segments, so as to improve the model's detection performance for road traffic accidents.

3 Traffic accident detection from the perspective of surveillance (TADS) dataset

This section details the collection and organization process of the TADS dataset, providing a detailed instruction on the categories of incidents, scenarios, weather conditions, lighting, and other accident attributes included in the dataset. The dataset is presented through a statistical analysis of the attributes, and a comparative analysis is conducted with several existing datasets.

3.1 Data collection

In order to construct a road traffic accident dataset of considerable scale from a surveillance perspective, we conducted searches on various online video platforms such as Youku, Tencent Video, and Bilibili using keywords such as "traffic accident" and "road accident." A total of 966 accident-related video segments were obtained, with a total duration of 145 min approximately, each containing an event of traffic accidents. The occurrence time of the traffic accidents within the video segment varied, potentially happening at the beginning, middle, or end of the segment. Subsequently, after obtaining these video clips, we utilized relevant software applications to extract

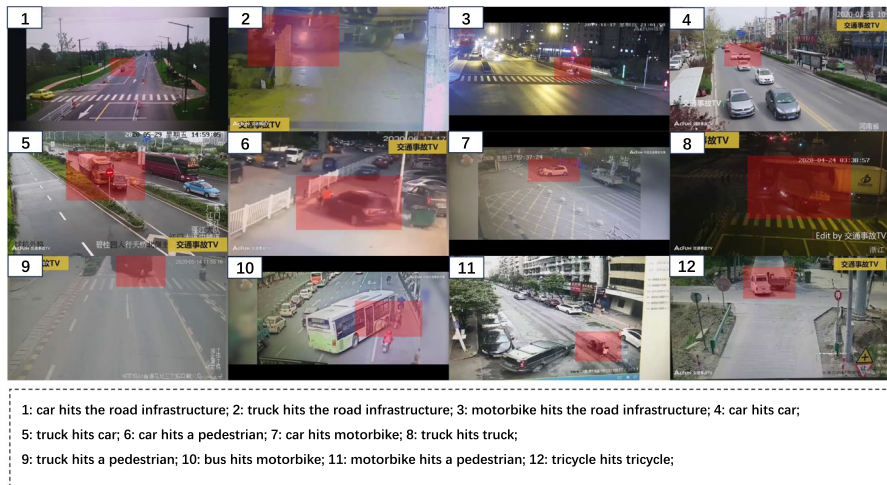


Fig. 1 Twelve accident categories and frame examples (red boxes cover accident areas in frames)

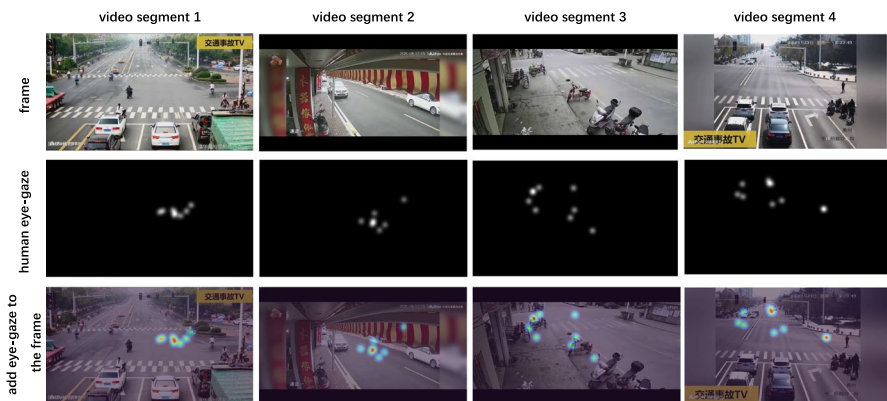


Fig. 2 Examples of normal video frames and their corresponding eye-gaze images

video frames, resulting in a total of 259,860 frames. Figure 1 displays some accident frame examples in the dataset.

Additionally, to obtain the gaze spatial annotations corresponding to each frame, we employed eye-tracking equipment to record the gaze areas of 10 human observers while watching the entire video segments in the dataset. The eye-tracking data were synchronized with the video frames, resulting in 259,860 annotated gaze images.

Figures 2 and 3 present some examples of normal frames and accident frames, along with their corresponding gaze images, respectively. From Figs. 2 and 3, it can be observed that during non-accident intervals (normal frame intervals), the attention areas in the gaze maps are relatively dispersed, whereas during the occurrence

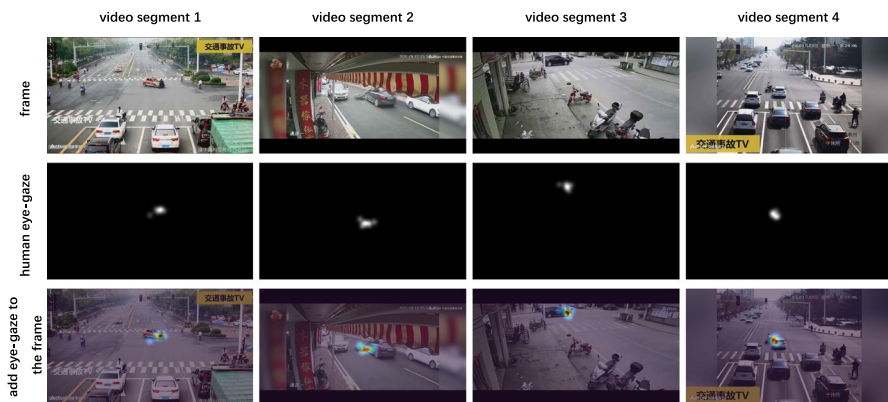


Fig. 3 Examples of accident video frames and their corresponding eye-gaze images

of accidents (accident frame intervals), the attention areas in the gaze maps are mostly concentrated on the accident regions within the frames. It is precisely due to this characteristic that we could use gaze images as spatial annotations for frames, to promote the understanding ability of traffic accidents and to enhance the performance of the model.

3.2 Accident attribute annotation

The TADS dataset is mainly intended to promote the research of road traffic accident detection from the surveillance perspective, so in order to facilitate subsequent comprehension and research on this issue, we extensively annotated the traffic accident attribute information contained in each video segment. Specifically, we annotated the attributes of each video segment as illustrated in Table 2.

It is worth noting that, in many cases, objects related to traffic accidents exhibit anomalous characteristics before the accidents actually occur. In order to more accurately annotate the start and end time of traffic accidents in video segments, refer to DADA-2000, TADS has also added annotations for abnormal starting frame positions before the accidents occur. An annotation example is shown in Fig. 4. According to the definitions of the start and end timestamps of traffic anomalies in Fig. 4, the TADS dataset contains 99,398 abnormal video frames and 160,493 normal video frames, totaling 259,891 video frames.

The statistical characteristics of scenery, position, and weather attributes in the TADS dataset are illustrated in a), b), and c) in Fig. 5, respectively.

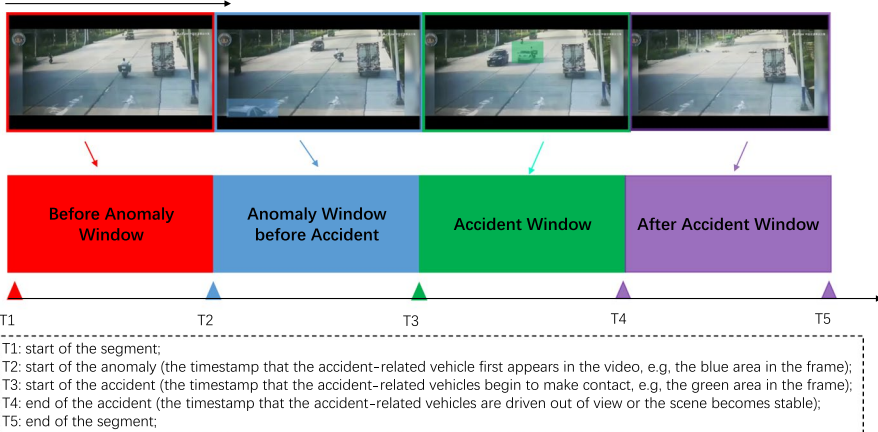
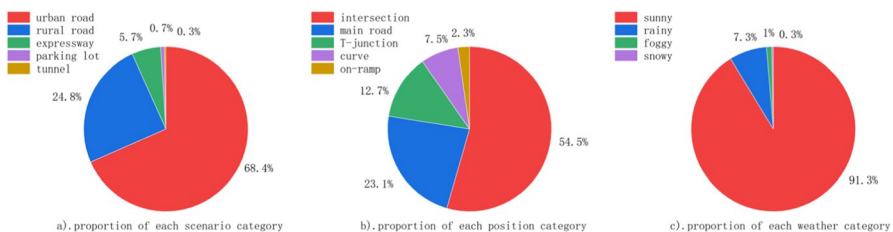
3.3 Accident categories

Due to the rich variety of road traffic accident categories within the dataset, following the methodology of DADA-2000, TADS initially categorizes

Table 2 Attributes annotated for each video segment

Attributes name	Detail
Accident number	A unique number to each video segment contained in the TADS dataset
Abnormal start frame	The location of the frame in which the accident-related vehicle first appears in the video or begins to exhibit unusual movement characteristics
Accident start frame	The location of the frame from which the accident-related vehicle began to make contact
Accident end frame	The location of frames in which the accident-related vehicle is driven out of view due to hitting, losing of control, or no longer has the characteristics of the accident motion
Accident categories	Divided into 12 categories according to the type of the entities involved in the accident; for details, see Sect. 3.3
Weather	Divided into four types: sunny, rainy, snowy, and foggy according to the weather conditions when the accident happened
Scenario	Divided into five types: expressway, urban road, rural road, tunnel, and parking lot according to the type of street scene when the accident happened
Position	Divided into five types: main road, curve, intersection, T-junction, and on-ramp according to the location of the street when the accident happened

accident categories: B-E(car hits motorbike); weather: sunny; scenario: rural road; position: main road;
time

**Fig. 4** An example of accident attribute annotation**Fig. 5** Statistical characteristics of scenery, position, and weather attributes in the TADS dataset

accident-involved entities into five types: road infrastructure (A), small vehicles (B), large vehicles (C), pedestrians (D), and small-scale rideable vehicles (E), as illustrated in Fig. 6.

According to the classification of accident-involved entities as depicted in Fig. 6, each category of accidents is assigned a unique identifier based on the actual situations of accidents appearing in the dataset. The correspondence between these identifiers and the entities involved in the accidents (collisions), along with their respective quantities, is detailed in Table 3, which presents the accident identifiers, the accident-involved entities, and their corresponding quantities.

3.4 Training and testing sets splitting

Considering the different motion characteristics of vehicles across different categories of accidents potentially, efforts were made to ensure a balanced distribution of the quantities of accidents in each category during the splitting of the training and testing sets. Specifically, to mitigate the impact of accident categories on detection performance of the model, a ratio of approximately 8:2 was employed for setting the sizes of the training and testing sets. This resulted in a proportional presence of each accident category in both the training and the testing sets, maintaining an 8:2 proportional relationship. The concrete quantities of each accident category in the training and testing sets, along with the total quantities, are detailed in Table 3, and the quantitative proportion of each category in the training and testing sets is illustrated in Fig. 7.

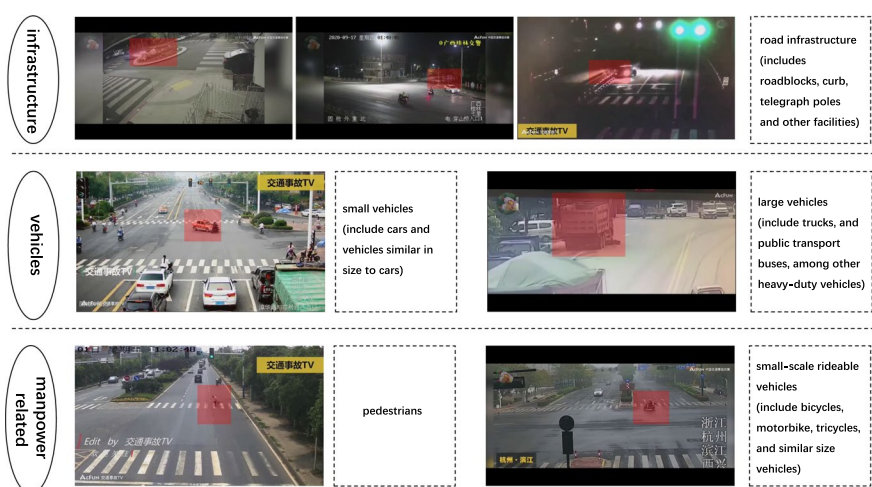
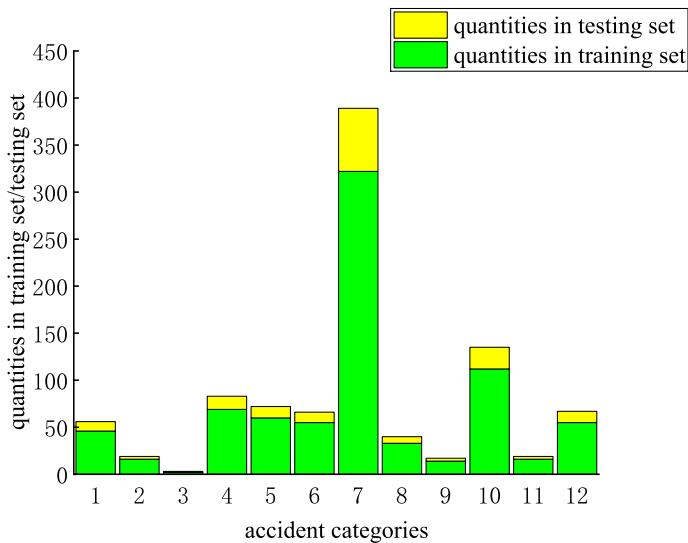


Fig. 6 The classification of entities relevant to accident categories in the TADS dataset

Table 3 Accident category identifier, their respective entities involved in accidents, and corresponding quantities

Category identifier	Entities involved ¹	Quantities in training set	Quantities in testing set	Total
1	A-B	46	10	56
2	A-C	16	3	19
3	A-E	2	1	3
4	B-B	69	14	83
5	B-C	60	12	72
6	B-D	55	11	66
7	B-E	322	67	389
8	C-C	33	7	40
9	C-D	14	3	17
10	C-E	112	23	135
11	D-E	16	3	19
12	E-E	55	12	67
Total	12	800	166	966

¹ “-” means “hit.” TADS classifies accidents into 12 categories based on the five types of accident-involved entities (from A to E), see Sect. 3.3

**Fig. 7** Each accident category and its quantity in the training and testing sets

3.5 Comparing with other typical datasets related

The comparison between the TADS dataset and the aforementioned typical datasets is presented in Table 4.

Table 4 Comparison between the TADS dataset and the aforementioned typical datasets

Datasets	Total duration	Accident ratio ¹	Perspective ²	Annotation amount	Annotations attribute ³
TAD [14]	About 1.2 h	261/333	S	24810 frames	S, T
UCF-Crime [34]	–	151/1900	S	1900 video segments	T
CADP [19]	About 5.2 h	1416/1416	S	205 video segments	S, T
DoTA [13]	–	4677/4677	F	4766 video segments	S, T
DADA-2000 [35]	–	2000/2000	F	2000 video segments	S, T
TADS	About 2.5 h	966/966	S	966 video segments	S, T

¹ accident ratio: accident segments quantity/total segments quantity.

²perspective: “S” means “the surveillance perspective;” “F” means “the first-person perspective.”

³annotations attribute: “S” means “spatial;” “T” means “temporal.”

From Table 4, it can be observed that both the DoTA and the DADA-2000 datasets possess sufficiently large scales and rich annotation information (temporal and spatial annotations). However, both are oriented toward the driver’s perspective (the first-person view). The CADP dataset is also with a large scale and is oriented toward the surveillance perspective (the third-person view), but the proportion of annotated part in the dataset is very small. Similarly, the TAD dataset, which is oriented toward the surveillance perspective (the third-person view), also lacks sufficient data volume and annotation quantity. In UCF-Crime, only a small portion belongs to abnormal categories related to road traffic accidents, and it only provides temporal annotations for the locations of abnormal occurrences. Therefore, considering factors such as dataset volume, annotation information, and perspective orientation, our dataset shows a certain superiority compared to the existing datasets above.

4 Methods

In this section, we construct a detection model RF-RG based on the special data elements in the TADS dataset referring to existing research on video anomaly detection or road traffic accident detection and then conduct a comparative analysis of the detection performance against prominent existing methods.

4.1 Model construction

Compared to popular accident datasets captured from the driver’s perspective (the first-person view), the proportion of the spatial area with accident occurrence within datasets captured from a surveillance perspective (the third-person view) is often smaller. These frames under the surveillance perspective often encompass larger areas of fixed monitoring backgrounds, which are typically unrelated to accident information. Therefore, following the approach outlined in [33], we preprocess the

The model utilizes U-Net as the architecture backbone, taking the RGB and optical flow values of four consecutive frames containing important objects from before the abnormal starting timestamp in the training set (normal frames) as input and producing predictions of the fifth frame’s RGB and gaze values. Since real-time

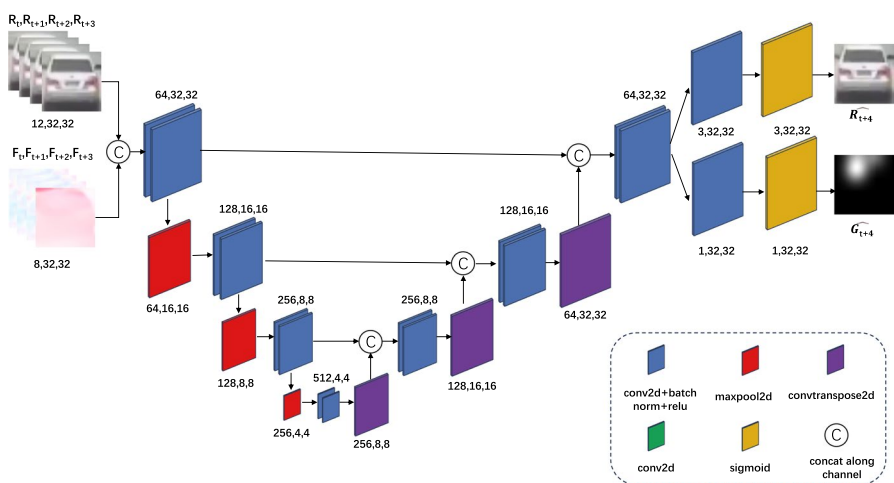


Fig. 8 The pipelines of the RF-RG model

gaze values corresponding to input data are not available in the real environment and differ from the training phase, during testing, only the mean squared error (MSE) of the RGB values between the predicted frame and the ground truth is used to compute the anomaly (accident) score for the predicted frame.

4.2 Loss function and accident discrimination

The model outputs two prediction values during training: $\hat{R}_{t+4}$ and $\hat{G}_{t+4}$. Consequently, the training loss considers two components: loss-rgb and loss-gaze. Loss-rgb calculates the mean squared error (MSE) between the predicted values $\hat{R}_{t+4}$ and the ground truth R_{t+4} at the pixel level, while loss-gaze incorporates Kullback–Leibler divergence (KLdiv), the normalized scanpath saliency (NSS), and the linear correlation coefficient (CC) losses between the predicted values $\hat{G}_{t+4}$ and the ground truth G_{t+4} which are commonly used in driving attention predicting task [12]. The formulation of the two kinds of losses can be expressed as follows:

$$\text{Loss-rgb} = \text{MSE}(R_{t+4}, \hat{R}_{t+4}) = \frac{(R_{t+4} - \hat{R}_{t+4})^2}{\sum i} \quad (1)$$

$$\begin{aligned} \text{Loss-gaze}(Y, \hat{Y}) = & \sum_i Y(i) \log \left(\epsilon + \frac{Y(i)}{\epsilon + \hat{Y}(i)} \right) - w_{cc} \frac{\text{cov}(Y, \hat{Y})}{\rho(Y)\rho(\hat{Y})} \\ & - w_{nss} \frac{1}{|P(i) \neq 0|} \sum_i \frac{Y(i) - \mu(Y)}{\rho(Y)} F(i) \end{aligned} \quad (2)$$

The index i spans across all pixels in the input or output, as illustrated in Fig. 8, where $\sum i = 32 \times 32 = 1024$. The loss term "loss-gaze" comprises three components: KLdiv, CC, and NSS, weighted by 1, w_{cc} , and w_{nss} , respectively. Therefore, the whole training loss can be defined as:

$$\text{Loss} = \text{Loss-rgb} + w_{gaze} \text{Loss-gaze} \quad (3)$$

where w_{gaze} also represents a weighting factor. During testing or in the real detection environment, the gaze data are unavailable, anomalies (incidents) in frame R_{t+4} are determined only based on the mean squared error (MSE) between the predicted value $\hat{R}_{t+4}$ and the ground truth R_{t+4} . So the anomaly (incident) score can be expressed as:

$$S = \text{norm} \left(\frac{\text{MSE}(R_{t+4}, \hat{R}_{t+4}) - \mu}{\sigma} \right) \quad (4)$$

In Eq. 4, μ and σ denote the mean and standard deviation of the mean squared errors (MSE) between the predicted values $\hat{R}_{t+4}$ and the ground truth R_{t+4} across all training sets, respectively, and $\text{norm}()$ denotes the popular 0 – 1 normalization operation. For frames where significant objects are not detected, their MSE values are replaced with 0. In cases where multiple significant objects are detected within a frame, the

maximum anomaly (incident) score is used to replace the anomaly (incident) score of that frame.

5 Experimental comparison and analysis

5.1 Evaluation metrics

The evaluation metrics for detection performance in this study include the Area Under Curve (AUC) and the detection accuracy.

AUC, representing the area under the Receiver Operating Characteristic (ROC) curve, is computed using the confusion matrix across all thresholds to obtain True Positive Rate (TPR) and False Positive Rate (FPR). A larger AUC signifies better detection performance. When the testing set consists of multiple video segments, as highlighted in [37], the AUC metric computed by concatenating all frames in the testing set is referred to as "micro-AUC," while averaging the AUC values obtained for each video segment is termed as "macro-AUC." Following the methodology described in [37], this paper calculates both types of AUC metrics for each model on the TADS dataset and compares them.

Additionally, as a binary classification task, accuracy measures the ratio of correctly detected frames to the total number of frames in the testing set when the detection threshold is set to 0.5, expressed as:

$$accuracy = \frac{\text{number of correctly detected frames}}{\text{total number of frames in the testing set}} \quad (5)$$

5.2 Experimental setup

All experiments conducted in this paper are based on the Ubuntu 20.04 system, utilizing the PyTorch framework, and powered by an Nvidia GeForce GTX 3090 GPU. The input and output image sizes during the experiments are set to 32×32 , with a batch size of 128. The optimization is performed using the Adam optimizer with a StepLR learning rate decay strategy, starting with an initial learning rate of $1e-4$. The total training epochs are set to 30, and the weight w_{gaze} in the loss function is assigned a value of 0.01.

For training, this experiment utilizes frames before the abnormal starting timestamp (normal frames), resulting in a training set comprising 56,823 frames. After extracting important objects and forming sequences of consecutive five frames, the training set consists of 178,468 sequences. The testing set comprises 39,707 frames, and after extracting important objects, it consists of 114,593 sequences. Due to the prevalent use of micro-AUC over macro-AUC in video anomaly detection, the model corresponding to the best micro-AUC across the 30 epochs is considered the optimal model.

5.3 Performance comparison

5.3.1 Typical methods

Classification based on supervised learning For the task of road traffic accident detection in videos, the ultimate goal is to accurately determine whether a frame belongs to an accident scene. Hence, from a supervised learning perspective, a binary classification model is trained to detect accidents on the testing set. This paper implements two typical supervised learning approaches: ResNet50, which extracts features using ResNet50 and then employs fully connected layers for binary classification, and LS-ResNet50, which first extracts spatial features for each frame in a group using ResNet50 and then feeds the corresponding features of each frame into an LSTM layer for temporal feature extraction, followed by binary classification using fully connected layers. It is noteworthy that in such methods, all 220,184 frames in the training set can be used for training.

Methods based on unsupervised video anomaly detection Due to the similarity between road traffic accidents and video anomalies, methods based on video anomaly detection could widely be used for road traffic accident detection currently. Similar to our approach, these methods primarily utilize autoencoders as the reconstruction or prediction models, training on normal video frames, and detecting anomalies based on reconstruction or prediction errors in the testing set. This paper compares the following video anomaly detection methods with our proposed approach.

STAE [32]: Autoencoders are effective means and commonly employed in anomaly detection. In reference [32], the authors focus on video tasks, leveraging the temporal and spatial feature extraction capabilities of ConvLSTM. They implement stacked autoencoders based on ConvLSTM for video anomaly detection.

MNAD [36]: Building upon autoencoders, this method introduces a memory mechanism into the reconstruction or prediction model to better memorize normal patterns in the training set, thus enhancing the reconstruction or prediction errors for abnormal frames in the testing set. The MNAD model used U-Net as the encoding–decoding reconstruction model, incorporating a memory module to improve anomaly detection capability. Note that by modifying the input, this method can perform anomaly detection in a predictive manner. In the original work [36], the predictive manner exhibits stronger anomaly detection capabilities compared to the reconstruction manner. Therefore, this paper compares the performance of this model in both manners.

OLP-TAD [31]: Similarly, employing U-Net as the encoding–decoding architecture, this method considers two losses of pixel intensity and pixel gradient regarding the spatial characteristics of frames. In the temporal domain, it incorporates optical flow data. Additionally, besides constructing a U-Net-based generator, this method builds a discriminator model, training in a Generative Adversarial Network (GAN) fashion.

5.3.2 Performance comparison and analysis

Overall performance comparison: After training, the performance of each model on the TADS testing set is presented in Table 5.

The following points can be inferred from the results above: The two kinds of AUC scores of the anomaly detection models used in the experiments are all > 0.5 (except the macro-AUC of the LS-ResNet50 model) on the TADS dataset, indicating there exists differences in distribution between abnormal (accident) frames and normal frames in this dataset, suggesting its suitability for studying road traffic accident detection under surveillance perspectives. Moreover, after training, these models can detect the abnormal (accident) frames in the dataset to some extent. In supervised learning classification methods, after considering temporal features, the LS-ResNet50 model outperforms the ResNet50 model to a large extent in terms of micro-AUC and accuracy metrics. In methods based on unsupervised video anomaly detection, the MNAD(pred) model, which detects anomalies through frame prediction mode, also outperforms the MNAD(recon) model, which detects anomalies through frame reconstruction mode, to a large extent in several metrics. This result is consistent with the experiment on Ped2 [27], Avenue [28], and ShanghaiTech [30] benchmark of MNAD in its original paper [36] on video anomaly detection. From the perspective of the most commonly used micro-AUC metric, our model achieves the best performance among several comparative models. We attribute this to the fact that, in addition to utilizing RGB and optical flow information similar to the several comparative models, our model extracts important objects in the frames in the preprocessing stage using popular object detection methods, thereby suppressing the influence of mostly irrelevant background in frames under surveillance perspectives to some extent. Furthermore, the eye-gaze data in the training set of TADS further aid in improving the model's performance. However, at the same time, compared to the detection performance on other non-traffic scene video anomaly detection benchmarks (such as MNAD(pred) with performances of 97.0 and 88.5 on Ped2 and Avenue datasets, respectively; OLP-TAD with performances of 95.4 and 85.1, respectively), the AUC or the accuracy values of these methods on the TADS dataset are significantly lower. We believe that the primary reason for the generally lower

Table 5 The performance of various models on the TADS testing set. Numbers in bold indicate the best results

Models	Evaluation metrics (%)		
	Micro-AUC	Macro-AUC	Accuracy
ResNet50	52.7	50.3	55.4
LS-ResNet50	59.0	47.2	61.8
STAE	56.9	59.6	60.4
MNAD(recon)	54.1	58.0	53.0
MNAD(pred)	66.8	66.6	63.7
OLP-TAD	61.3	63.7	63.2
RF-RG	68.6	69.1	62.8

performance of models on the TADS dataset lies in the challenging nature of the dataset itself. The TADS dataset is a new accident dataset that encompasses various scenarios, weather conditions, and locations, making it inherently complex. Therefore, conventional methods may not be sufficient to effectively deal with such a novel and intricate dataset, that is, indicating that there are still many challenges on the TADS dataset for road traffic accident detection tasks, so there is still much work to be done in the future.

Various categories of accidents performance comparison of representative methods: To provide a more in-depth comparison of the above methods, we selected the LS-ResNet50 model, which exhibited the best overall performance in the supervised comparative methods, and the MNAD(pred) model, which demonstrated the best overall performance in the unsupervised comparative methods. We then conducted separate comparisons with the RF-RG method on a test set of 12 classes of traffic accident data, as shown in Fig. 9.

From Fig. 9, it is evident that for accident categories 1, 2, 4, 6, 9, 10, 11, and 12, our method RF-RG yielded the best results among the three methods. For categories 5, 7, and 8, RF-RG approached the performance of the best method among the three. However, for category 3, RF-RG produced relatively poor results. We attribute this discrepancy in category 3 to potential differences in data volume. As observed in Fig. 7, category 3 had the lowest data volume in the entire dataset. Such a scarcity or imbalance in data volume can impact the model's learning process, leading to an insufficient understanding of the data patterns related to that particular accident category. This ultimately results in poorer detection outcomes for that category of accidents.

Upon analyzing the detection results across various accident categories, it is evident that RF-RG consistently achieves the best results for the majority of accident categories.

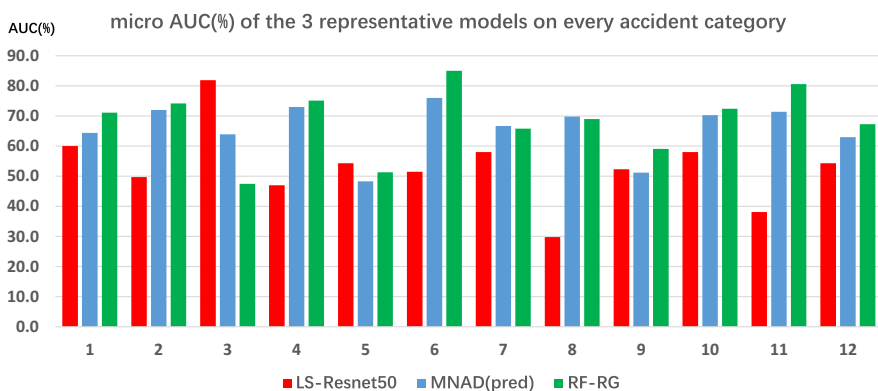


Fig. 9 Detection performance of the three representative models on various categories of accidents in the testing set

Table 6 The performance of different data components utilized in the TADS dataset. Numbers in bold indicate the best results

Models	Evaluation metrics (%)		
	Micro-AUC	Macro-AUC	Accuracy
r	65.8	67.0	63.1
r-f	68.0	68.8	63.0
r-f-g	68.6	69.1	62.8

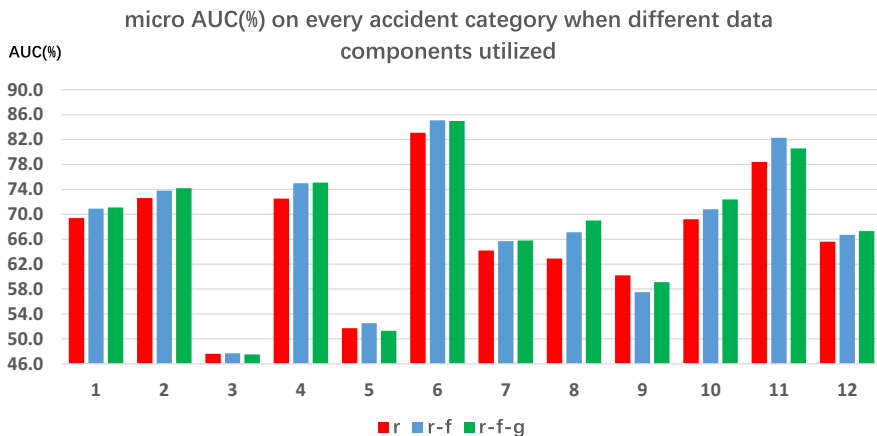


Fig. 10 Detection performance on various categories of accidents when different data components are utilized in the testing set

5.3.3 Ablation experiment

Overall performance comparison in ablation: To understand the impact of various data components contained in the TADS dataset on detection performance, we sequentially removed eye-gaze data and optical flow data from the model to compare performance changes, as shown in Table 6.

In this table, r, r-f, and r-f-g represent experiments conducted using only RGB values, both the RGB and optical flow values simultaneously, and the RGB values, optical flow, and corresponding gaze values of the frames simultaneously. From the perspective of the most commonly used micro-AUC metric, we can see that, after training, in addition to the frame's RGB values, incorporating optical flow values and gaze values correspondingly can sequentially improve the model's detection performance. Although the accuracy metrics have decreased slightly, the difference is minimal. From the perspective of the AUC metric, the optimal detection performance is achieved when all three data components are simultaneously utilized in the model.

Various categories of accidents performance comparison in ablation: Figure 10 presents a comparison of the detection performance of the model on the 12 accident categories in the testing set under three conditions: using only RGB data, using both RGB and optical flow data, and using RGB and optical flow data

along with gaze data simultaneously from TADS dataset during training. It can be observed from the figure that, in most cases of the 12 accident categories, the detection performance is significantly better when utilizing RGB, optical flow, and gaze data simultaneously (categories 1, 2, 4, 7, 8, 10, and 12) compared to using RGB data solely or RGB data combined with optical flow data. The difference is minimal for category 6. For category 3, all three approaches exhibit the lowest performance, possibly due to the limited presence of category 3 in the TADS dataset (only three instances). As for these few categories that do not conform to our expected patterns, such as category 9 and category 11, we believe there may be two reasons for this. Firstly, we attribute this to the challenging nature of the dataset. The TADS dataset contains a variety of scenarios, weather conditions, and locations data related to the accidents, and the inherent complexity of the dataset poses a certain challenge for the task of accident detection. Conventional methods, such as detecting accidents based on frame reconstruction errors, may not effectively address challenging accident detection scenarios. Secondly, the amount of data associated with these minority categories is relatively low within the entire dataset. For example, category 9 and category 11, as shown in Fig. 7, contain significantly less data compared to other categories. The limited amount of data may lead the model not to fully learn the characteristics of accidents in these categories, thus failing to produce the expected results. In conclusion, the comparison of detection performance for each category indicates that among most accident categories, each data component contained in the TADS dataset can contribute to accident detection effectively.

6 Conclusion

In response to the current scarcity of datasets for road traffic accident detection from a surveillance perspective characterized by small-scale and insufficient annotation, this paper proposes a novel dataset named TADS (Traffic Accident Detection from the Perspective of Surveillance). Compared to existing datasets, TADS exhibits certain superiority in terms of total frames, total accident segments, categorization of accidents, and annotation richness. Additionally, TADS dataset contains gaze data in equal quantity with the frames counts as the spatial annotation to aid in anomaly (accident) detection-related learning. Leveraging the special data components in the TADS dataset, a new detection model is constructed based on mainstream video anomaly detection methods, and its performance is compared with major supervised classification methods and video anomaly detection methods on the TADS dataset. Experimental results indicate the suitability of TADS dataset for research on road traffic accident detection from a surveillance perspective, with its data components contributing to enhancing model detection performance. However, there remains significant room for improvement in detection performance on the TADS dataset. Therefore, numerous related studies can still be pursued in the future to address this issue better.

Author contributions For this research, C.Y., J.F., and W.S. designed the concept of the research; L.H. collected the original video frames; C.Y. conducted data organization and annotation, designed the models, implemented experimental design, conducted the data analysis, and wrote the draft paper; W.S. and J.F. reviewed the whole paper.

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Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Authors and Affiliations

Yachuang Chai¹ · Jianwu Fang² · Haoquan Liang³ · Wushouer Silamu^{1,4}

✉ Jianwu Fang
j.w.fangit@gmail.com

Yachuang Chai
cyc@stu.xju.edu.cn

Haoquan Liang
lianghaoquan490@gmail.com

Wushouer Silamu
wushour@xju.edu.cn

¹ School of Computer Science and Technology, Xinjiang University, Shengli Road 666, Urumqi 830046, Xinjiang, China

² College of Artificial Intelligence, Xi'an Jiaotong University, Xianning West Road 28, Xi'an 710049, Shaanxi, China

³ School of Electronics and Control Engineering, Chang'an University, Wenyi Road 126, Xi'an 710064, Shaanxi, China

⁴ Xinjiang Key Laboratory of Multilingual Information Technology, Shengli Road 666, Urumqi 830046, Xinjiang, China